

Byung-Kwan Lee

Ph.D., School of Electrical Engineering at KAIST

Research Scientist at  NVIDIA

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 LinkedIn ·  Github ·  Google Scholar ·  Huggingface

Research Interest

Building efficient, high-performing Vision-Language Models (VLMs), with focus on:

- Distillation
- RL
- Agent
- Pruning

Work Experience

NVIDIA • Research Scientist

Oct. 2025 - Current

LEAD ZPPO: Zone of Proximal Policy Optimization [COMPLETED] RL post-training that keeps the teacher inside the prompt rather than the policy gradient, using BCQ/NCQ question reformulations and a prompt replay buffer to lift small students on hard questions without policy drift.

Under Review U.S. Provisional Patent Filed 2026 [ArXiv Release](#) Internal Release

LEAD AXPO: Agent eXplorative Policy Optimization [COMPLETED] Agentic RL training that recovers tool usage through tool-call resampling, improving multimodal reasoning performance against larger baselines.

Under Review U.S. Provisional Patent Filed 2026 [ArXiv Release](#) Internal Release [Internal Tech Transfer](#)

LEAD Masking Teacher and Reinforcing Student [COMPLETED] Mask-progressive RL distillation that gradually un.masks teacher weights and uses offline RL with accuracy & distillation rewards.

CVPR 2026 Accept U.S. Non-Provisional Patent Filed 2026 [ArXiv Release](#) Internal Release

LEAD Unified RL & Imitation Learning for VLMs [COMPLETED] Combines RL with adversarial imitation, using an LLM-based discriminator and multi-teacher guidance to build lightweight yet powerful VLMs.

NeurIPS 2025 Accept U.S. Patent Upgraded to Non-Provisional 2025 [ArXiv Release](#) Internal Release

LEAD GenRecal [COMPLETED] Cross-architecture VLM distillation via a Recalibrator that aligns heterogeneous token representations regardless of vocabulary, token splits, or index ordering.

ECCV 2026 Accept U.S. Patent Upgraded to Non-Provisional 2025 [Internal Tech Transfer](#)

LEAD VLsI: Verbalized Layers-to-Interactions [COMPLETED] Hardened the verbalizer-based layer-wise distillation into a production-ready recipe and led its internal tech transfer into NVIDIA's VLM development pipeline.

[Internal Tech Transfer](#)

NVIDIA • Research Intern

Oct. 2024 - Oct. 2025

LEAD Unified RL & Imitation Learning for VLMs [INITIATED] First formulation of the RL + adversarial imitation training pipeline; initial experiments and team setup.

U.S. Provisional Patent Filed 2025

LEAD GenRecal [INITIATED] Initial design of the Recalibrator framework for cross-tokenizer VLM distillation; first proof-of-concept and internal demo.

U.S. Provisional Patent Filed 2025 [ArXiv Release](#) Internal Release

LEAD VLsI: Verbalized Layers-to-Interactions [INITIATED] Layer-wise distillation using intermediate verbalizers, enabling small VLMs (2B/7B) to align with large VLMs' reasoning progression and outperform GPT-4V.

CVPR 2025 Accept U.S. Non-Provisional Patent Filed 2025 [ArXiv Release](#) Internal Release

Education

KAIST

Ph.D., School of Electrical Engineering

◦ GPA: 3.77/4.3

◦ Dissertation: *Building High-performing, Efficient-size Vision Language Models: Merge, Modify, and Distill* [Link]

Mar. 2020 - Aug. 2025

KAIST

M.S., The Cho Chun Sik Graduate School of Green Transportation

◦ GPA: 3.72/4.3

◦ Thesis: *Training Encoder-Attention through Fully-Connected CRFs for Efficient End-to-End Lane Detection Model* [Link]

Mar. 2018 - Feb. 2020

Hanyang University

B.S., Mathematics and Electronic Engineering

◦ GPA: 3.86/4.5

◦ Thesis: *Learning Rank of Evaluated Songs for Classifying Unknown Music by Convolutional Neural Network* [Link]

Mar. 2014 - Feb. 2018

Publications

Overall 18 Accepts · 4 Pending · 2 Tech Reports

CV 5 CVPR · 2 ICCV · 3 ECCV · 1 ICIP

ML 3 NeurIPS · 1 ICLR

NLP 1 ACL · 1 EMNLP

Journal 1 Pattern Recognition



■ 79% (19/24) First & Lead Author
■ 21% (5/24) Co-Author

1. Zone of Proximal Policy Optimization: Teacher in Prompts, Not Gradients

Byung-Kwan Lee, Ximing Lu, Shizhe Diao, Minki Kang, Saurav Muralidharan, Karan Sapra, Andrew Tao, Pavlo Molchanov, Yejin Choi, Yu-Chiang Frank Wang, Ryo Hachiuma

Under Review [Paper][Project]

U.S. Patent Application Filed (Provisional), NVIDIA Research, 2026

2. SpatialClaw: Rethinking Action Interface for Agentic Spatial Reasoning

Seokju Cho, Ryo Hachiuma, Abhishek Badki, Hang Su, Byung-Kwan Lee, Chan Hee Song, Sifei Liu, Subhashree Radhakrishnan, Seungryong Kim, Yu-Chiang Frank Wang, Min-Hung Chen

Under Review [Paper][Project][Code]

U.S. Patent Application Filed (Provisional), NVIDIA Research, 2026

3. Agent eXplorative Policy Optimization for Multimodal Agentic Reasoning

Minki Kang, Shizhe Diao, Ryo Hachiuma, Sung Ju Hwang, Pavlo Molchanov, Yu-Chiang Frank Wang, Byung-Kwan Lee

Under Review [Paper][Project]

U.S. Patent Application Filed (Provisional), NVIDIA Research, 2026

4. Hide to See: Reasoning-prefix Masking for Visual-anchored Thinking in VLM Distillation

Seonghoon Yu, Dongjun Nam, Byung-Kwan Lee[†], Jeany Son[†]

Under Review [Paper][Code]

5. Why and When Visual Token Pruning Fails? A Study on Relevant Visual Information Shift in MLLMs Decoding

Jiwan Kim, Kibum Kim, Wonjoong Kim, Byung-Kwan Lee, Chanyoung Park

European Conference on Computer Vision (ECCV), 2026 [Paper][Project]

6. GenRecal: Generation after Recalibration from Large to Small Vision Language Models

Byung-Kwan Lee, Ryo Hachiuma, Yong Man Ro, Yu-Chiang Frank Wang, and Yueh-Hua Wu

European Conference on Computer Vision (ECCV), 2026 [Paper][Project]

U.S. Patent Application Filed (Non-Provisional), NVIDIA Research, 2025

- 7. Masking Teacher and Reinforcing Student for Distilling Vision-Language Models**
Byung-Kwan Lee, Yu-Chiang Frank Wang, Ryo Hachiuma
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026 [[Paper](#)]
U.S. Patent Application Filed (Non-Provisional), NVIDIA Research, 2026
- 8. Recursive Think-Answer Process for LLMs and VLMs**
Byung-Kwan Lee*, Youngchae Chee*, Yong Man Ro
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings, 2026 [[Paper](#)][[Project](#)]
- 9. RefineBench: Evaluating Refinement Capability in Language Models**
Young-Jun Lee*, Seungone Kim*, Byung-Kwan Lee, Minkyong Moon, Yechan Hwang, Jong Myoung Kim, Graham Neubig, Sean Welleck, Ho-Jin Choi
International Conference on Learning Representations (ICLR), 2026 [[Paper](#)][[Project](#)]
Best Runner-Up Award, Workshop for Multi-Turn Interactions in Large Language Models, NeurIPS 2025 (Oral, Top 1%) [[Link](#)]
- 10. Unified Reinforcement and Imitation Learning for Vision-Language Models**
Byung-Kwan Lee, Ryo Hachiuma, Yong Man Ro, Yu-Chiang Frank Wang, Yueh-Hua Wu
Neural Information Processing Systems (NeurIPS), 2025 [[Paper](#)][[Project](#)]
U.S. Patent Application Filed (Non-Provisional), NVIDIA Research, 2025
- 11. MultiVerse: A Multi-Turn Conversation Benchmark for Evaluating Large Vision and Language Models**
Young-Jun Lee, Byung-Kwan Lee, Jianshu Zhang, Yechan Hwang, Byungsoo Ko, Han-Gyu Kim, Dongyu Yao, Xuankun Rong, Eojin Joo, Seung-Ho Han, Bowon Ko, Ho-Jin Choi
IEEE/CVF International Conference on Computer Vision (ICCV), 2025 [[Paper](#)][[Project](#)]
Workshop for Knowledge-Intensive Multimodal Reasoning, ICCV 2025 [[Link](#)]
- 12. VLsI: Verbalized Layers-to-Interactions from Large to Small Vision Language Models**
Byung-Kwan Lee, Ryo Hachiuma, Yu-Chiang Frank Wang, Yong Man Ro[†], Yueh-Hua Wu[†]
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025 [[Paper](#)][[Project](#)]
U.S. Patent Application Filed (Non-Provisional), NVIDIA Research, 2025
- 13. Phantom of Latent for Large Language and Vision Models**
Byung-Kwan Lee, Sangyun Chung, Chae Won Kim, Beomchan Park, Yong Man Ro
Technical Report [[Paper](#)][[Code](#)][[Huggingface Model](#)]
- 14. TroL: Traversal of Layers for Large Language and Vision Models**
Byung-Kwan Lee, Sangyun Chung, Chae Won Kim, Beomchan Park, Yong Man Ro
Empirical Methods in Natural Language Processing (EMNLP), 2024 [[Paper](#)][[Code](#)][[Huggingface Model](#)]
- 15. Meteor: Mamba-based Traversal of Rationale for Large Language and Vision Models**
Byung-Kwan Lee, Chae Won Kim, Beomchan Park, Yong Man Ro
Neural Information Processing Systems (NeurIPS), 2024 [[Paper](#)][[Code](#)][[Huggingface Model](#)]
31st Samsung HumanTech Paper Awards in section of Computer Science & Engineering
- 16. MoAI: Mixture of All Intelligence for Large Language and Vision Models**
Byung-Kwan Lee, Beomchan Park, Chae Won Kim, Yong Man Ro
European Conference on Computer Vision (ECCV), 2024, [[Paper](#)][[Code](#)][[Huggingface Model](#)]
- 17. CoLLaVO: Crayon Large Language and Vision mOdel**
Byung-Kwan Lee, Beomchan Park, Chae Won Kim, Yong Man Ro
Findings of the Association for Computational Linguistics (ACL), 2024, [[Paper](#)][[Code](#)][[Huggingface Model](#)]
2024 KCC XAI Workshop, Best Paper Awards

18. **Causal Unsupervised Semantic Segmentation**
Junho Kim*, Byung-Kwan Lee*, and Yong Man Ro
Journal of Pattern Recognition [[Paper](#)][[Code](#)]
19. **Mitigating Adversarial Vulnerability through Causal Parameter Estimation by Adversarial Double Machine Learning**
Byung-Kwan Lee*, Junho Kim*, and Yong Man Ro
IEEE/CVF International Conference on Computer Vision (ICCV), 2023 [[Paper](#)][[Code](#)]
20. **Mitigating Dataset Bias in Image Captioning through CLIP Confounder-free Captioning Network**
YeonJu Kim, Junho Kim, Byung-Kwan Lee, Sebin Shin, and Yong Man Ro
IEEE International Conference on Image Processing (ICIP), 2023 [[Paper](#)][[Code](#)]
21. **Demystifying Causal Features on Adversarial Examples and Causal Inoculation for Robust Network by Adversarial Instrumental Variable Regression**
Junho Kim*, Byung-Kwan Lee*, and Yong Man Ro
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023 [[Paper](#)][[Code](#)]
22. **Masking Adversarial Damage: Finding Adversarial Saliency for Robust and Sparse Network**
Byung-Kwan Lee*, Junho Kim*, and Yong Man Ro
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022 [[Paper](#)][[Code](#)]
23. **Distilling Robust and Non-Robust Features in Adversarial Examples by Information Bottleneck**
Junho Kim*, Byung-Kwan Lee*, and Yong Man Ro
Neural Information Processing Systems (NeurIPS), 2021 [[Paper](#)][[Code](#)]
24. **Towards Adversarial Robustness of Bayesian Neural Network through Hierarchical Variational Inference**
Byung-Kwan Lee*, Youngjun Yu, and Yong Man Ro
Technical Report [[Paper](#)][[Code](#)]

Reviewer Experience

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)	Journal
IEEE Transactions on Neural Networks and Learning Systems (TNNLS)	Journal
IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)	Journal
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)	Conference
IEEE/CVF International Conference on Computer Vision (ICCV)	Conference
European Conference on Computer Vision (ECCV)	Conference
International Conference on Learning Representations (ICLR)	Conference
Neural Information Processing Systems (NeurIPS)	Conference
International Conference on Machine Learning (ICML)	Conference
AAAI Conference on Artificial Intelligence (AAAI)	Conference
Association for Computational Linguistic (ACL)	Conference
Empirical Methods in Natural Language Processing (EMNLP)	Conference

Invited Talks and Awards

Silver Reviewer Award, International Conference on Machine Learning (ICML)	2026
Best Runner-Up Award (Oral, Top 1%), Multi-Turn Interactions in LLMs Workshop @ NeurIPS	2025
Invited Talk in Kongju University	2025
Invited Talk in Hanyang University ERICA	2025
Invited Talk in Kookmin University	2025
Invited Talk in Dongseo University	2025

31st Samsung HumanTech Paper Awards in section of Computer Science & Engineering	2025
KCC XAI Workshop, Best Paper Awards	2024
Invited Talk in Korea Institute of Science and Technology Information (KISTI)	2024
Invited Talk in NAVER HyperCLOVA X	2024
Invited Talk in Kakao Brain	2024
1st Award for Research Presentation in Center for Applied Research in Artificial Intelligence (CARAI)	2023
GIRE Research Mentor, Seoul International School – “MUsE”, Journal of Student Research [Publication]	2023
KAIST SW IT Academy, AI Project Director of 1st Award Topic	2023
Invitation to Presenter in Korean Conference on Computer Vision (KCCV)	2023
KAIST SW IT Academy, Lecturer of Data Structure & Algorithm, Deep Learning, Computer Vision	2023
Invitation to Presenter in Korean Conference on Computer Vision (KCCV)	2022
2nd Award for Research Presentation in Center for Applied Research in Artificial Intelligence (CARAI)	2022
KAIST Fellowship	Mar. 2020 - Feb. 2024
National Government Fellowship	Mar. 2018 - Feb. 2020
1st President’s Award for International Autonomous Vehicle Competition, Team Leader [YouTube][Talk]	2018